Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 41

Rooted and Spanning Trees

Covers Chartrand & Zhang, Section 13.2

Status of this set

Ungraded practice. Solutions released with it. **The last practice set of the course.**

Part I: How to Practice This Material

The three facts

1. A rooted tree's vocabulary is all defined by **distance from the root**.
2. A connected graph of order n and size m needs $m - n + 1$ edges removed to leave a spanning tree.
3. A rooted tree where every vertex has at most c children and height h has order at most $1 + c + c^2 + \cdots + c^h$.

Three mistakes to expect

1. **Confusing height with order.** Height counts levels below the root.
2. **Off-by-one on levels.** The root is level 0, so height h means $h + 1$ levels.
3. **Computing $m - n$ instead of $m - n + 1$.**

Part II: Practice Problems

Section A — Rooted trees

1. A tree has root v_1 and edges $v_1v_2,\ v_1v_3,\ v_2v_4,\ v_3v_5,\ v_3v_6,\ v_5v_7$.

(a) List the vertices at each level, and give the height.

(b) Give the parent of v_7 and the children of v_3.

(c) List the leaves and the ancestors of v_7.

2.

(a) What is the maximum order of a binary tree of height 5?

(b) A rooted tree has every vertex with at most 3 children and height 4. What is its maximum order?

(c) A binary tree has 1000 vertices. What is the least possible height?

Section B — Spanning trees

3. For each connected graph, give the size of a spanning tree and the number of edges that must be removed.

(a) Order 15, size 28.

(b) K_7.

(c) A 5-regular graph of order 12.

4. Let G be the graph on $V = \{1, \dots, 5\}$ with $E = \{12, 13, 23, 34, 35, 45\}$.

(a) How many edges must be removed to leave a spanning tree?

(b) Give one spanning tree by listing its edges.

(c) Does G have more than one spanning tree? Explain without listing them all.

Section C — Writing

5. In four to six complete sentences, explain why a graph has a spanning tree if and only if it is connected, and describe a procedure for finding one.